Decision file: Unit 47 product cooling and storage (DEXPI reference P&ID C01)

Status: DRAFT for engineering review, not approved. Prepared 2026-10-29 11:13 +03 by PlantAgent from the working model.

Question: Sales wants +20% throughput (18.0 to 21.6 m3/h). Can the unit do it? Which equipment limits first, and what is the cheapest fix that works?

Result at 120 % of today's throughput (re-solved in DWSIM for this file): all stated limits within 100 %.

1. What changes and why

Rev	Change	Reason	Proposed by	Applied
Rev 1	Cooling water to H1007 flow: 20 → 27 m³/h	decision-file test: Raise H1007 cooling water to 27 m3/h	muse-spark-1.3-contributor	2026-10-29T11:12:12+0300
Rev 2	Chilled water to H1008 flow: 15 → 18 m³/h	decision-file test: Raise H1008 chilled water to 18 m3/h	muse-spark-1.3-contributor	2026-10-29T11:12:35+0300
Rev 3	T4750 tank working volume: 20.8 → 22.1 m³	decision-file test: Raise T4750 working level to 22.1 m3	muse-spark-1.3-contributor	2026-10-29T11:13:00+0300

2. Limits before and after, at 120 % (DWSIM, re-solved now)

Equipment or stream	Check	Before	After	Value after	Limit
H1007 heat exchanger	area needed for hot outlet 35 C	106 %	81 %	37.76 m2	46.8 m2
P4712 outlet (product leaving the unit)	temperature vs design	103 %	96 %	23.9 C	25 C
T4750 tank	residence time	103 %	97 %	62.09 min	60 min
H1007 outlet to T4750	temperature vs design	101 %	96 %	33.5 C	35 C
P4711 pump	head vs rated	88 %	88 %	10.54 m	12 m
P4712 pump	flow vs rated	88 %	88 %	43.86 m3/h	50 m3/h
P4711 pump	flow vs rated	86 %	86 %	21.6 m3/h	25 m3/h
Chilled water to H1008	flow vs 90 % of available supply	83 %	100 %	18.0 m3/h	18.0 m3/h
P4712 pump	shaft power vs motor	81 %	81 %	6.09 kW	7.5 kW
H1007 heat exchanger	installed area vs frame	78 %	78 %	46.8 m2	60 m2
Cooling water to H1007	flow vs 90 % of available supply	74 %	100 %	27.0 m3/h	27.0 m3/h
H1007 heat exchanger	cold-side pressure drop vs available	50 %	86 %	0.858 bar	1.0 bar

3. Drawing

Redlined P&ID: `redline.svg` (revision clouds at H1007, H1008, T4750).

4. Relief protection (P&ID relief scan)

• H1007 MN side: no relief on this P&ID. no relief device reachable Heated side that can be blocked in.

• H1007 WK side: no relief on this P&ID. no relief device reachable (lines leave the drawing: check the connecting P&ID) Heated side that can be blocked in.

• H1008 MN side: isolatable. relief 47125/SpringLoadedGlobeSafetyValve only through 47126/C7: if one is closed the side has no relief Heated side that can be blocked in.

• H1008 QS side: no relief on this P&ID. no relief device reachable (lines leave the drawing: check the connecting P&ID) Heated side that can be blocked in.

• T4750 MN side: no relief on this P&ID. no relief device reachable

• T4750 ? side: no relief on this P&ID. no relief device reachable

• The changes are operating changes; the relief scan reads the drawing, so its findings above are unchanged by them and still need closing.

5. Field checks and assumptions

• No field checks were raised in the answer.

• Assumption (model): Site datasheets supersede P&ID where stated: P4711 rated 25 m3/h at 12 m, motor 2.2 kW, min 8 m3/h, NPSHr 2.5 m used; P&ID 200 m3/h/10 m/60 kW is template error per P&ID data_quality warning.

• Assumption (model): P4712 P&ID 420 m3/h/84 kW is same template error; site max stroke 50 m3/h + motor 7.5 kW used, P&ID DesignPressureHead 40 m retained (no site head given), PSV set 6 barg from P&ID valve set_pressure + site confirmation.

• Assumption (model): T4750 P&ID DN20000x4 m (=1257 m3) inconsistent with 26 m3 nominal per data_quality warning; site nominal 26 m3 used, working volume 20.8 m3 at current 80% level per Current operation row (tank volume_m3 is working volume).

• Assumption (model): H1007/H1008 U=600 W/m2K fouled 2024 test used at current flows (U_ref [18,20] and [22.5,15] m3/h); clean 900 W/m2K not used - normal fouled operation.

• Assumption (model): H1007/H1008 installed area 46.8 m2 from P&ID DesignHeatTransferArea; H1007 frame limit 60 m2, CW available dP 1.0 bar and supply limits 30/20 m3/h (90% planning margin enforced by simulator) from site datasheets.

• Assumption (model): Product 7 mol% ethanol modelled as Water 0.93 / Ethanol 0.07 mole, NRTL package for polar liquids.

• Assumption (model): T4750 modelled as well-mixed tank at 1.013 bar; two inlets (H1007 product + H1008 recirc) combined in MIX_TANK mixer upstream (DWSIM tank has one inlet); residence_basis sDispatch (product leaving for good) per 60-min QC requirement.

• Assumption (model): Discharge branch modelled as splitter fixing recirc branch 22020 kg/h (=22.5 m3/h) letting dispatch take rest, reproducing stroke/recirc operations; PSV line 47125 excluded (standby relief, check_coverage relief_path, no normal flow).

• Assumption (model): 47124 suction trim and 47126 isolation ball valves lumped as fully open (no dP); only flow-carrying control valves 47123/C1, 47127/C1, 47141/C1 modelled explicitly with P&ID tags.

• Assumption (model): No H1008 dP_ref given in data sheet so none imposed (never invent); 47141/C1 dP 0.3 bar assumed small outlet loss to close chilled-water pressure balance (utility return not measured).

6. Second opinion (Muse Spark Open, answered independently)

Status: **partly**.

• Differs: limiting equipment: H1007, H1008, T4750 vs H1007, T4750

• Differs: feeds.s47140.m3_h: only the main model changes it

• Differs: verified in DWSIM: H1007, H1008, T4750 vs H1007, T4750

• Agrees: feeds.s47130.m3_h: 27.0 vs 27.0

• Agrees: units.T4750.volume_m3: 22.1 vs 22.1

• Agrees: reaches the target: yes vs yes

7. Pre-startup safety review (PSSR) checklist (generated from the changes)

• [ ] Verify the cooling water to h1007 can deliver 27 m³/h at header pressure, and that the branch control valve can pass it.

• [ ] Measure the pressure drop across H1007 at the new flow (model: cold side 0.858 bar, hot side 0.417 bar) and compare with the available differential.

• [ ] Confirm the cooling water to h1007 return temperature stays within the utility system's limit at the new duty.

• [ ] Confirm H1007's relief protection is unchanged by the new duty (see the relief scan section).

• [ ] Verify the chilled water to h1008 can deliver 18 m³/h at header pressure, and that the branch control valve can pass it.

• [ ] Confirm the chilled water to h1008 return temperature stays within the utility system's limit at the new duty.

• [ ] Confirm H1008's relief protection is unchanged by the new duty (see the relief scan section).

• [ ] Confirm the high-level alarm and trip on T4750 leave margin above the new working volume of 22.1 m³; update the level setpoint in the operating procedure.

• [ ] Update the P&ID, equipment datasheets and the operating procedure for every change above.

• [ ] Brief the operators on the new setpoints before start-up.

• [ ] Record the first 24 h of operation at the new throughput and compare with the model's predicted values in this file.

8. Sign-off (DRAFT: not approved)

Role	Name	Signature	Date
Process engineer			
Operations			
Process safety (MOC)			
Approver			

9. Reproducibility

• Model: v2-calibrated built by muse-spark-1.3-contributor on 2026-10-28 15:46; P&ID coverage missing connections: 0

• Audit-chain head hash: `14220a889ff646d443f2f9662a090331bce0bfcc79cca5e3d3c403243ffce743` (chain intact: True)

• Code version: 648a410; simulator: DWSIM (headless) via plantagent.process

• Values in sections 2 and 7 were recomputed when this file was made; re-run `python -m plantagent.decision chem_c01` on the same revisions to reproduce them.